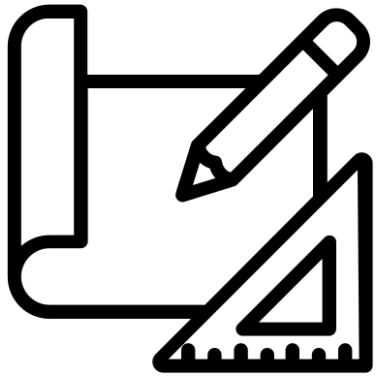


INTRODUCTION to BLOCKCHAIN

CHAPTER2: TECHNICAL ARCHITECTURE AND BASIC COMPONENTS

Dr. Noureddine Lasla

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Chapter Overview

Objective:

Understanding the building blocks of blockchain systems.

Key Areas of Focus:

- Introduction to Blockchain Architecture
- Blockchain Architecture (Nodes, Ledgers, P2P Net.)
- Block Structure (Merkle Trees and Hash Functions)
- Public and Private Key Cryptography
- Data Integrity, Security Mechanisms, and Attacks

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Introduction to Blockchain Architecture

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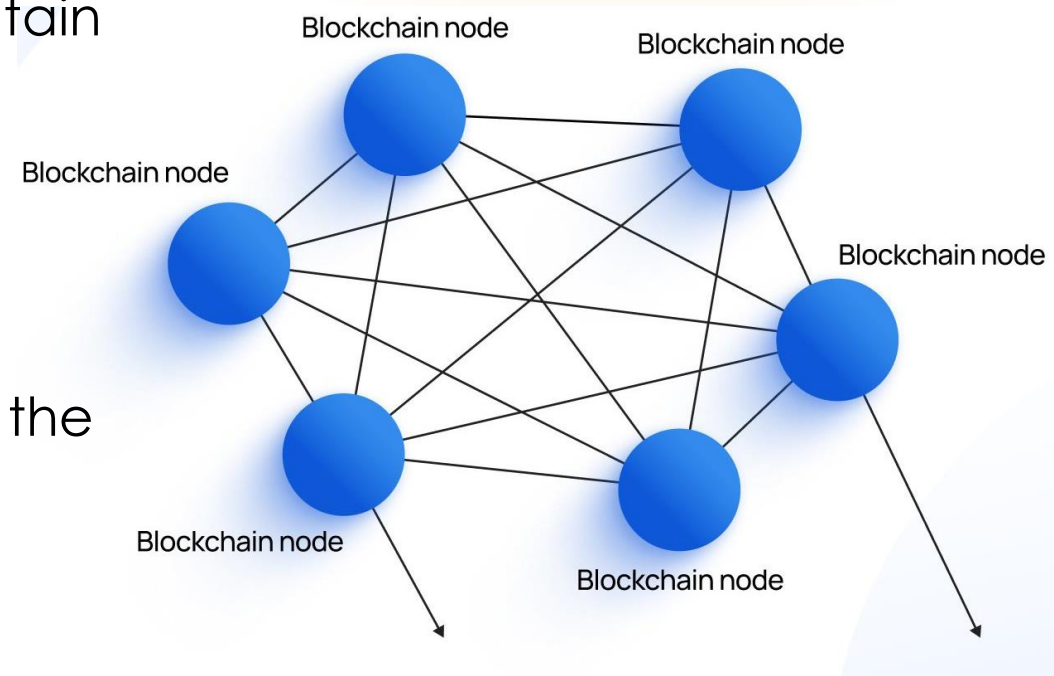
What is Blockchain Architecture?

- Blockchain architecture refers to the structural framework and design principles that underpin a decentralized and distributed digital ledger system.
- It enables trustless interactions between participants without the need for intermediaries.
- This architecture is built upon key components such as nodes, cryptographic hashing, consensus mechanisms, and a peer-to-peer network,

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Key Components of Blockchain Architecture

1. **Blockchain Nodes:**
   - Independent devices (computers) that maintain the blockchain's ledger.
   - Store and validate data in a decentralized manner.
   - Types:
     - **Full nodes:** maintain a complete copy of the blockchain and participate in validating transactions),
     - **Light nodes** only store the most recent headers

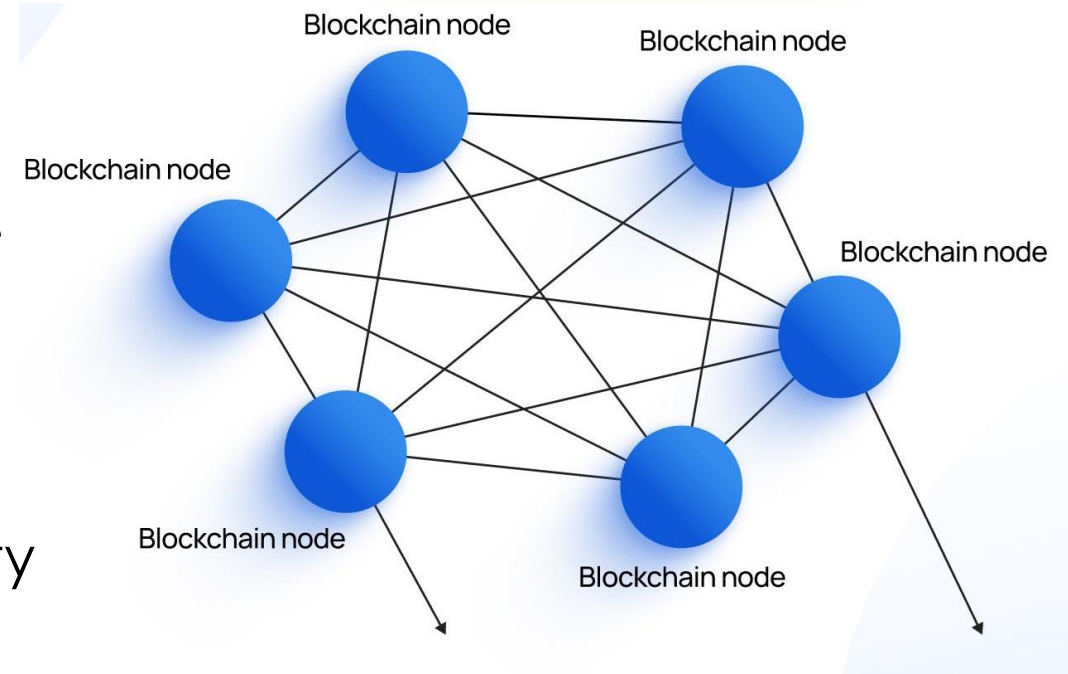


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Key Components of Blockchain Architecture

Blockchain Nodes Functionalities:

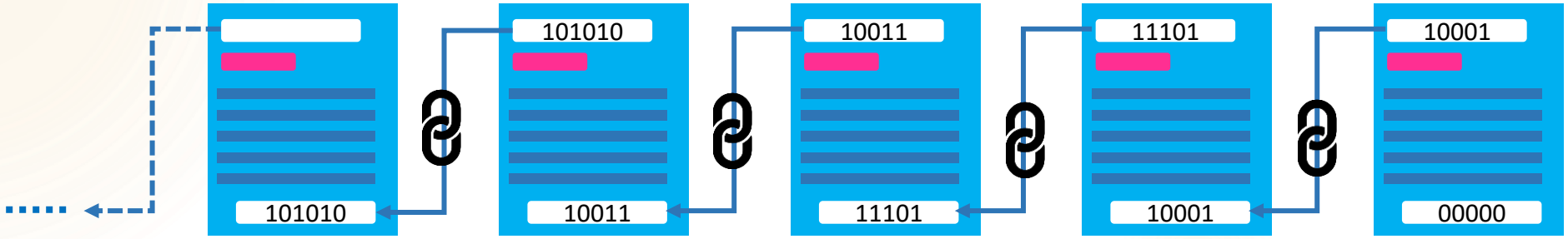
- Maintenance and synchronization of ledger copies
  - If few individual nodes are affected, the data remains safe and accessible for the network as a whole
- Transaction validation and consensus mechanism
- User interaction and network connectivity
  - Block explorers.



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Key Components of Blockchain Architecture

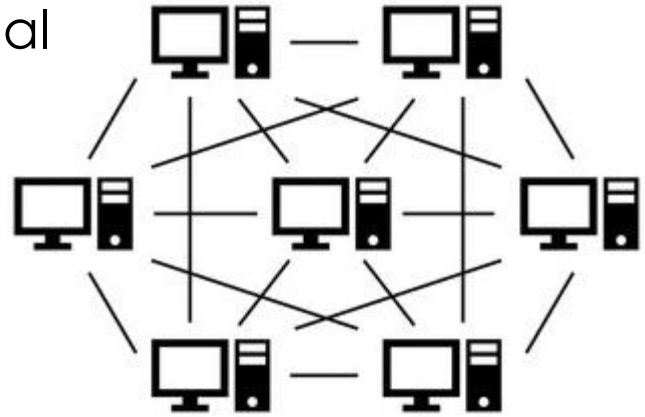
2. **Ledger**
   - Set of blocks chained together
     - Each block is a units of data containing transactions.
   - A distributed record of all transactions within the network
   - Shared and synchronized among all nodes in the blockchain system.



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Key Components of Blockchain Architecture

3. **Peer-to-Peer Network**
   - Allows blockchain nodes to directly interact with each other
   - Nodes use the P2P network to exchange information, such as: transactions, block proposals, and consensus messages.
   - Ensures that updates (e.g., new transactions or blocks) are distributed across all nodes efficiently and securely.
   - Enhances fault tolerance since the system remains operational even if some nodes go offline.



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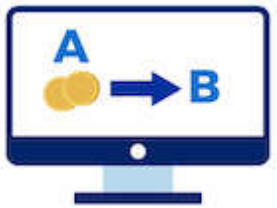
Key Components of Blockchain Architecture

4. **Consensus Mechanisms**
   - Algorithms that ensure agreement across nodes (e.g., Proof of Work, Proof of Stake).
   - Prevents double-spending and ensures consistency.



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Blockchain Transaction Lifecycle



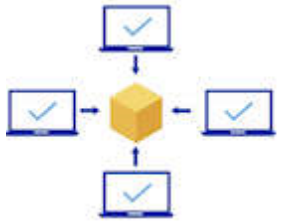
A **transaction** is sent by user to the net.



A new block is created to include that **transaction**



The new block is broadcasted to the entire net.



Nodes in the net. Validate the block and **transaction**



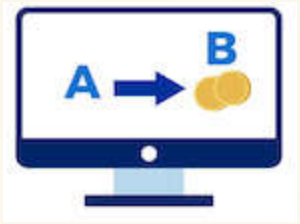
Nodes received reward for their participation (PoW)



The new block is added to the blockchain ledger



The blockchain update is distributed across the net.



The **transaction** is complete

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Block Structure

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Inside the Block Header

Fields in the Header:

- Hash of the Previous Block:
  - Links the current block to the previous one.
  - Ensures immutability.
- Merkle Root:
  - Root hash of the Merkle tree (represents transactions).
  - Efficient verification of data integrity.
- Timestamp
- Nonce
- Difficulty Target:
  - Defines the mining challenge level.

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Hash Function



A function that converts input data into a fixed-size alphanumeric string. Example: SHA-256 (Secure Hash Algorithm).

- **Key Properties:**
  1. Deterministic: Same input always produces the same output.
  2. Collision-Resistant: Two different inputs cannot produce the same output.
  3. Fast Computation: Hashes are computed efficiently.
  4. Pre-image Resistance: Cannot reverse-engineer the input from the hash.
- **Role in Blockchain**
  - Secures transaction data.
  - Links blocks using hash pointers.
  - Ensures tamper-proof records.

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Hash Function



Example:

SHA-256("Blockchain") =

4a8a08f09d37b73795649038408b5f33fba404ba622bfbb23b6ad13c6b5b1e1c

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Hash Function in Practice



- **SHA-256 (Secure Hash Algorithm 256-bit):**
  - Widely used in Bitcoin for Proof of Work (PoW).
  - Strong cryptographic security.
- **Keccak-256 (SHA-3):**
  - Used in Ethereum.
  - Resistant to length-extension attacks.

Hash Function	Key Application	Strengths	Use in Blockchain
SHA-256	Mining, PoW	Secure, fast	Bitcoin
Keccak-256	Smart Contracts	Resistant to attacks	Ethereum

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Merkle Tree



A binary tree where leaf nodes represent hashes of transactions, and parent nodes represent combined hashes of their children.

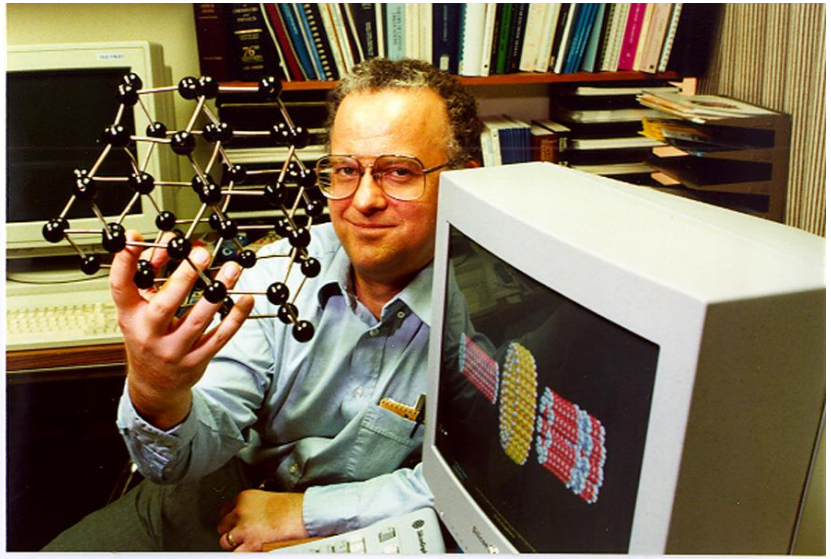
Structure

1. Each transaction is hashed individually
2. Hash pairs are combined iteratively to create the Merkle root

- **Benefits**
  - Efficient transaction verification
  - Reduced storage requirements.
  - Ensures integrity without downloading the entire blockchain.

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Merkle Tree



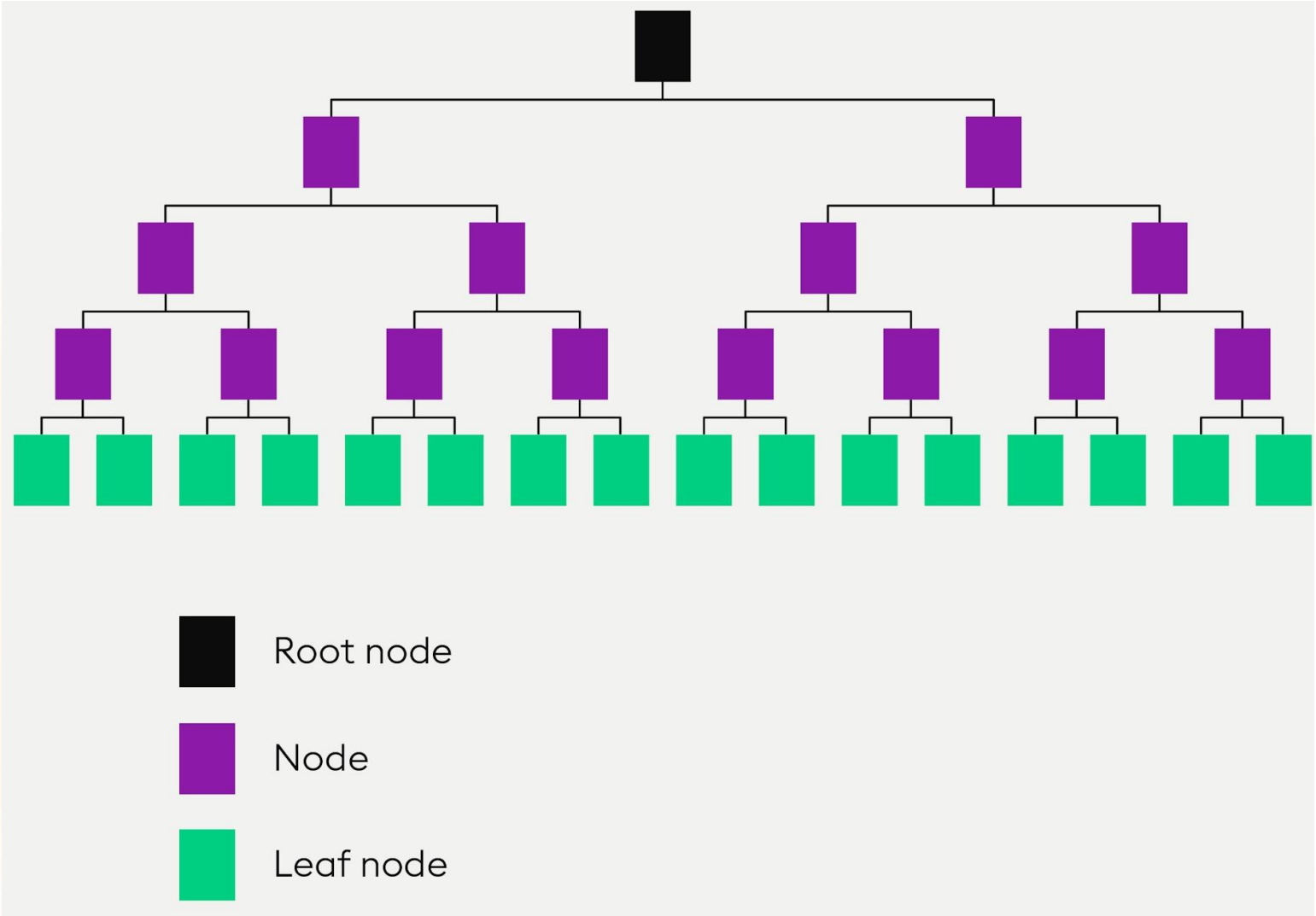
Ralph Merkle patented the **hash tree** in 1979

Is an American computer scientist and mathematician. He is one of the inventors of public-key cryptography and the inventor of cryptographic hashing,

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Merkle Trees



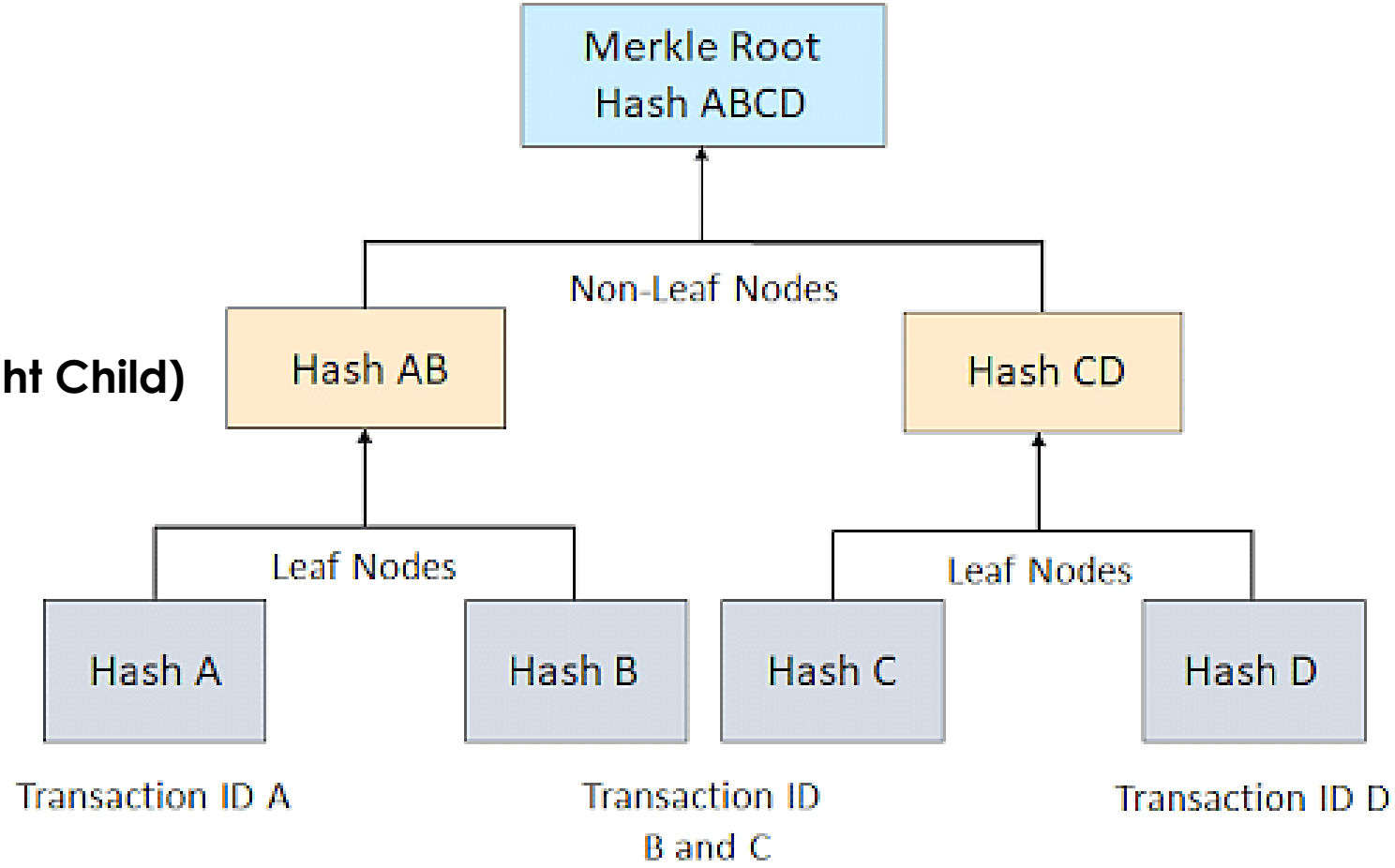


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Merkle Trees



Hash(Node) = Hash(Left Child || Right Child)

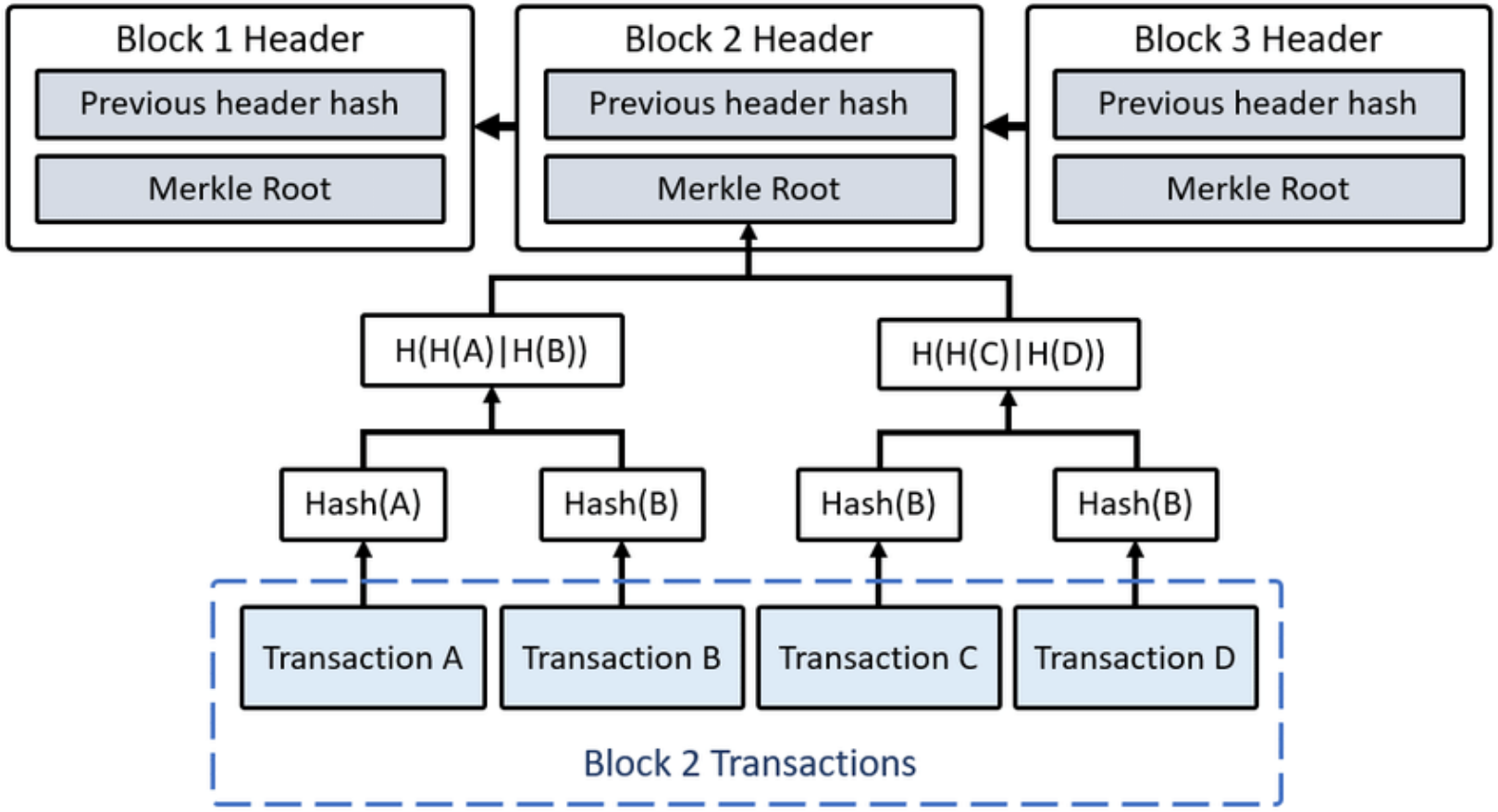


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Merkle Trees



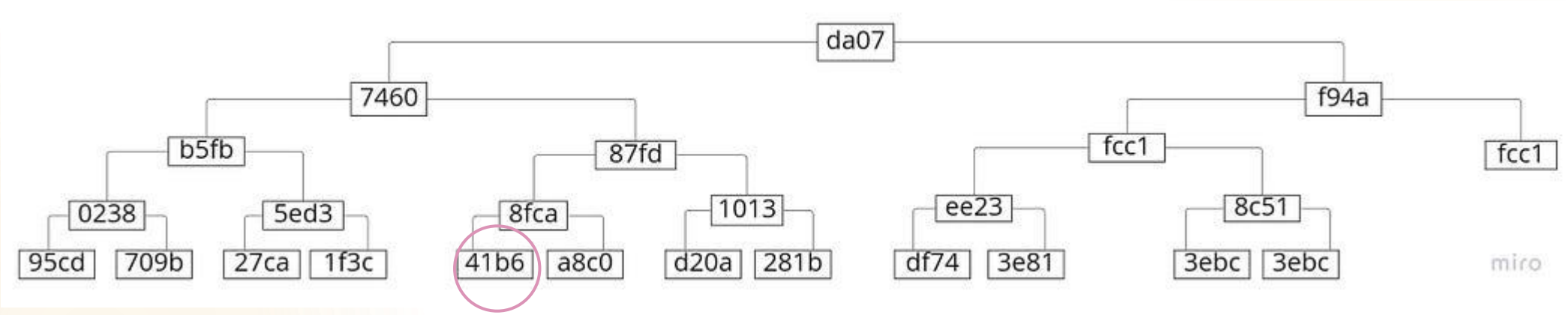
The root hash summarizes all data contained in related individual transactions



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Merkle Trees

Merkle tree example :

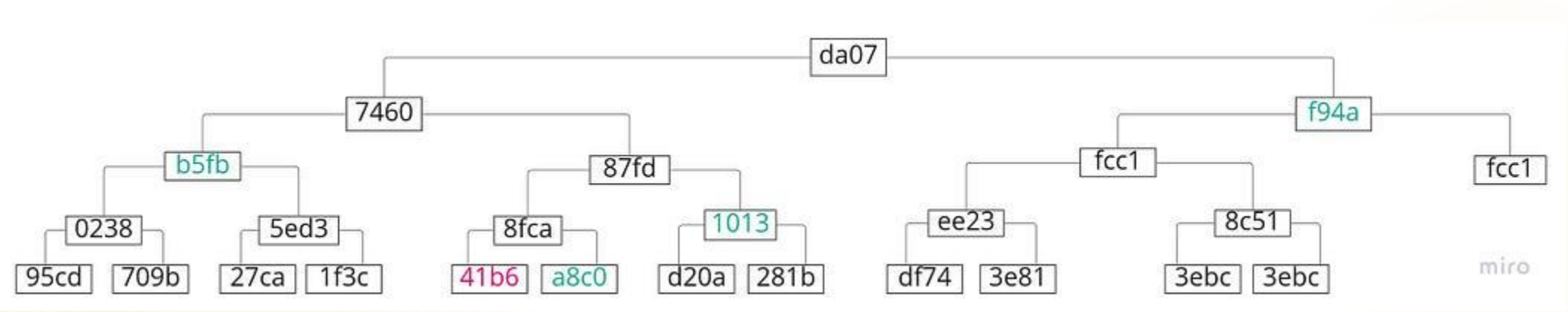


Want to Proof that 41b6 is actually included in the Merkle tree

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Merkle Trees

only need to have the hashes in color: [41b6, a8c0, 1013, b5fb, f94a]



With ***41b6***, we need ***a8c0*** to reconstruct *8fca*.
With *8fca*, we need ***1013*** to reconstruct *87fd*.
With *87fd*, we need ***b5fb*** to reconstruct *7460*.
With *7460* we need ***f94a*** to reconstruct ***da07***, the Merkle root.

If after applying our hash (***41b6***) to the Merkle proof we get the expected ***da07*** Merkle root, then we know that our hash is part of the Merkle root.

we can determine that we only need about log(n) number of hashes to be able to check if a hash belongs to a certain Merkle root, which is efficient

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Benefits of Merkle Trees in Blockchain

- **Scalability:**
  - Enables light clients (e.g., SPV - Simplified Payment Verification) to verify transactions without downloading the entire blockchain.
- **Verification Efficiency:**
  - Verifying a transaction only requires a few hashes along the Merkle path (logarithmic time complexity).
- **Tamper Detection:**
  - If any transaction or data changes, the Merkle root changes, enabling quick detection of tampering.
- **Storage Optimization:**
  - Transactions are summarized into a single hash, reducing the space required in the block header.

Introduction to Cryptography in Blockchain

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Introduction to Cryptography

Cryptography secures communication and ensures data **integrity** and **authenticity**.

Types of Cryptography in Blockchain:

- **Symmetric Cryptography:** Single key for encryption and decryption (not commonly used in blockchain).
- **Asymmetric Cryptography:** Public and private key pair for secure communication (widely used).

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Introduction to Cryptography

- **Importance in Blockchain:**
  - Secures transactions.
  - Provides user identity through digital signatures.
  - Protects the integrity of the blockchain.

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Public and Private Keys in Blockchain

- Key Components:
  - Public Key: Shared openly, used for encryption.
  - Private Key: Kept secret, used for decryption and digital signatures.
- How It Works:
  - Sender encrypts data with the recipient's public key.
  - Recipient decrypts the data using their private key.
- Use in Blockchain:
  - Public keys are used as **wallet addresses.**
  - Private keys **sign** transactions to **prove authenticity**.

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Digital Signatures in Blockchain



Is a cryptographic technique that ensures:

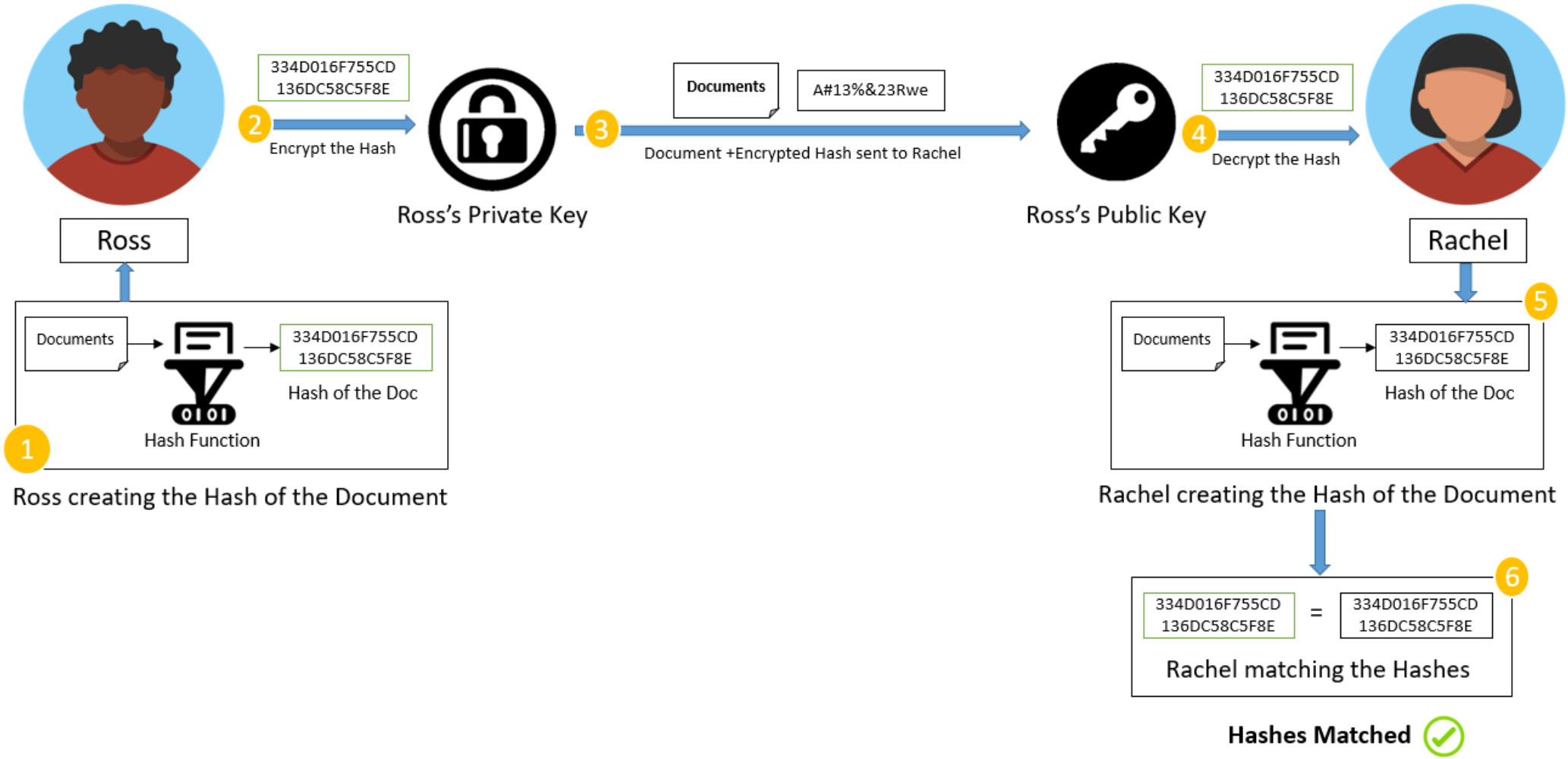
- Authenticity: Verifies the sender.
- Integrity: Confirms the data has not been altered.

- How It Works:
  - Sender signs the data using their private key.
  - Receiver verifies the signature using the sender's public key.
- Use in Blockchain:
  - **Validates** transactions.
  - **Prevents** unauthorized access.

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Digital Signatures in Blockchain





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Security Mechanisms in Blockchain

Ensuring Security in Blockchain

- **Consensus Mechanisms:**
  - Proof of Work (PoW).
  - Proof of Stake (PoS).
  - Byzantine Fault Tolerance (BFT).
- **Encryption:**
  - Public and Private Key Cryptography.
  - Secures transaction data.
- **Smart Contract Audits:**
  - Prevents vulnerabilities in automated code execution

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Common Security Threats to Blockchain

- 51% Attack:
  - Malicious actors gain majority control of network.
- Sybil Attack:
  - Attackers create multiple fake identities to disrupt consensus.
- Double Spending Attack:
    - Exploiting system to spend the same asset multiple times.
- Eclipse Attack:
  - Isolating nodes to manipulate their view of the blockchain.

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Preventing Blockchain Attacks

- Distributed Consensus Protocols:
  - Mitigates 51% attacks.
- Node Verification:
  - Prevents Sybil attacks by authenticating identities.
- Encryption Standards:
  - Secures data transfer and storage.
- Regular Network Audits:
  - Identifies vulnerabilities proactively.

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